A Simplified Simscape IGBT Model with Single-Point Switching Energies and Thermal Ports

Model documentation — `igbt_simple.ssc`, `igbt_simple_2th.ssc`

Abstract

This note documents a minimal Simscape language component that represents an IGBT with its antiparallel free-wheeling diode (FWD) for loss and thermal studies. Conduction is described by fixed voltage drops. Switching losses are described by three single-point energies, E_{on}, E_{off} and E_{rr}, given at one reference operating point and scaled linearly with the commutated current and the blocking voltage. The total dissipated power is delivered to one or two thermal conserving ports: a single-port version lumps the two chips, a two-port version separates the transistor and the diode so that the two junction temperatures can be evaluated independently. The document states the equations, the sampling scheme used to recover the commutated current and voltage, the energy-to-power conversion, the parameter extraction from the device datasheet, and the validity limits of the model. Section 8 documents the two-port version, including the local detection of the diode recovery event. Default parameters correspond to the Mitsubishi Electric CM1200DW-34T (1700 V, 1200 A half-bridge module).

1 Scope and modelling assumptions

The component is intended for system-level simulation of converters where the quantities of interest are the average losses and the junction temperature, not the shape of the switching transient. The following assumptions apply.

1. The switching transient is not resolved in time. Each commutation is treated as an instantaneous event that injects a finite amount of energy into the thermal path. No $\mathrm{d}i/\mathrm{d}t$, no tail current and no reverse recovery current are visible on the electrical terminals.

2. Conduction is represented by a fixed voltage drop. The output characteristic is therefore a vertical line in the I–V plane, not the datasheet curve.

3. All parameters are constant. In particular they do not depend on the junction temperature computed by the thermal network. The user is expected to enter the values corresponding to the expected operating temperature, typically 125 °C.

4. In the single-port version the IGBT and the FWD share one thermal port, so the model returns the total loss of one switch position and cannot separate the two junctions. The two-port version of Section 8 removes this restriction.

Assumption 3 removes the electro-thermal feedback loop. This is a deliberate simplification: it makes the model non-iterative and numerically robust, at the cost of a loss estimate that is correct only near the chosen temperature. If the simulated junction temperature departs significantly from the assumed one, the parameters must be re-entered and the simulation repeated.

2 Component interface

The port side declared in the trailing comment of a node declaration accepts `left` and `right`; `top` and `bottom` are not accepted for this component and cause a compilation error.

Table 1: Ports.

Name	Type	Domain	Meaning
`p`	conserving	electrical	collector
`n`	conserving	electrical	emitter
`H`	conserving	thermal	case/junction heat flow
`g`	input	physical signal	gate–emitter voltage, V
`P`	output	physical signal	total instantaneous loss, W

Table 2: Parameters and default values (CM1200DW-34T).

Symbol	Name	Default	Source
$V_{\mathrm{ce,sat}}$	`Vce_sat`	2.40 V	typ. at 1200 A, 125 °C, terminal
V_{f}	`Vf`	2.80 V	V_{EC} typ. at 1200 A, 125 °C
V_{th}	`Vth`	6 V	$V_{\mathrm{GE(th)}}$ typ.
V_{hyst}	`Vhyst`	0.85 V	user choice
$E_{\mathrm{on,ref}}$	`Eon_ref`	138 mJ	table value, see Sec. 7
$E_{\mathrm{off,ref}}$	`Eoff_ref`	309 mJ	table value, see Sec. 7
$E_{\mathrm{rr,ref}}$	`Err_ref`	220 mJ	table value, see Sec. 7
$I_{\mathrm{c,ref}}$	`Ic_ref`	1200 A	test condition
$V_{\mathrm{dc,ref}}$	`Vdc_ref`	1000 V	test condition
R_{on}	`Ron`	0.05 mΩ	numerical
R_{d}	`Rd`	0.05 mΩ	numerical
G_{off}	`Goff`	1 µS	numerical
τ_{s}	`tau_s`	1 µs	numerical
τ_{p}	`tau_p`	200 µs	numerical

3 Conduction

Let $v = v_{\mathrm{p}} - v_{\mathrm{n}}$ be the terminal voltage and i the terminal current flowing from `p` to `n`. The terminal current is split into a transistor current i_{t} and a diode current i_{d},

$$i = i_{\mathrm{t}} + i_{\mathrm{d}}, \tag{1}$$

each governed by a piecewise linear characteristic:

$$i_{\mathrm{t}} = \begin{cases} \dfrac{v - V_{\mathrm{ce,sat}}}{R_{\mathrm{on}}}, & g > V_{\mathrm{th}} \ \wedge \ v > V_{\mathrm{ce,sat}}, \\ G_{\mathrm{off}}\, v, & \text{otherwise}, \end{cases} \tag{2}$$

$$i_{\mathrm{d}} = \begin{cases} \dfrac{v + V_{\mathrm{f}}}{R_{\mathrm{d}}}, & v < -V_{\mathrm{f}}, \\ G_{\mathrm{off}}\, v, & \text{otherwise}. \end{cases} \tag{3}$$

The two series resistances are not intended as a physical model of the output characteristic. They exist to make each conduction mode an explicit, invertible relation between v and i: a strictly fixed voltage drop would impose $v = V_{\mathrm{ce,sat}}$ as an algebraic constraint independent of current, and the mode selection at current reversal would become implicit, which is a common source of chattering. With $R_{\mathrm{on}} = 0.05\,\mathrm{m\Omega}$ the extra drop at 1200 A is 60 mV, i.e. 2.5 % of $V_{\mathrm{ce,sat}}$.

If a more faithful conduction model is wanted, R_{on} and R_{d} can be set to the slope of the datasheet output characteristic and $V_{\mathrm{ce,sat}}$, V_{f} to the corresponding zero-current intercepts. This turns (2)–(3) into the usual $v = V_0 + R\,i$ representation at no extra cost.

The off-state conductance G_{off} closes the mesh when the device is blocking. Both branches contribute, so the leakage current at 1200 V is 2.4 mA and the associated blocking loss is 2.9 W. This is of the same order as the datasheet I_{CES} limit and is negligible with respect to the other loss terms; G_{off} may be reduced if a smaller residual is preferred, at the price of a larger resistance ratio and a stiffer network.

The gate input is compared with the threshold V_{th}. The model therefore accepts a gate signal expressed in volts, and a driver output swinging between $-V_{\text{GE}}$ and $+V_{\text{GE}}$ can be fed directly.

4 Switching losses

4.1 Scaling law

The three energies are given at a single reference point $(I_{\text{c,ref}}, V_{\text{dc,ref}})$ and scaled bilinearly:

$$E_x(I,V) = E_{x,\text{ref}} \, \frac{|I|}{I_{\text{c,ref}}} \, \frac{|V|}{V_{\text{dc,ref}}}, \qquad x \in \{\text{on}, \text{off}, \text{rr}\}. \tag{4}$$

The dependence on the blocking voltage is close to linear for a hard-switched IGBT, so the second factor is a good approximation over the usual DC-link range. The dependence on current is not exactly linear: the datasheet curves of E_{on} and E_{off} versus I_{C} are slightly convex, and E_{rr} saturates at high current. Equation (4) therefore over-estimates the energy at low current and under-estimates it near the rated current if the reference point is taken at rated current. Choosing the reference point near the expected average commutated current, rather than at the datasheet rated point, reduces this error.

4.2 Recovering the commutated current and voltage

The quantities to be inserted in (4) are the current that the device commutates and the voltage it blocks. Neither is available at the switching instant itself:

- at turn-on the pre-event voltage is the blocking voltage, but the pre-event current is zero;
- at turn-off the pre-event current is the conducted current, but the pre-event voltage is $V_{\text{ce,sat}}$.

Moreover, the value that the solver associates with a continuous variable inside an event equation is not guaranteed to be the pre-transition one, so sampling v and i directly in the `when` clause is unreliable.

The model avoids the problem by never sampling at the event. Two continuous first-order states track the required quantities during the intervals in which they are meaningful, and hold them otherwise:

$$\dot{i}_{\text{h}} = \begin{cases} \dfrac{i_{\text{t}} - i_{\text{h}}}{\tau_{\text{s}}}, & g > V_{\text{th}}, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

$$\dot{v}_{\text{h}} = \begin{cases} \dfrac{v - v_{\text{h}}}{\tau_{\text{s}}}, & g < V_{\text{th}} \ \wedge \ v > 0, \\ 0, & \text{otherwise}. \end{cases} \tag{6}$$

Because i_{h} and v_{h} are continuous states, they cannot jump at an event; whatever the solver does with v and i at the transition, the held values are the pre-transition ones. With $\tau_{\text{s}} \ll t_{\text{on,min}}$ the tracking error is negligible.

The scheme has a useful side effect. At turn-on, i_{h} contains the transistor current measured just before the *previous* turn-off. For $f_{\text{sw}} \gg f_{\text{out}}$ this is the load current to be commutated, which

is exactly what (4) requires. If instead the diode was conducting during the previous on-state, then $i_\mathrm{t} \approx 0$ was tracked and $i_\mathrm{h} \approx 0$, so the model assigns $E_\mathrm{on} \approx 0$. This is the correct behaviour: an IGBT turning on while its own antiparallel diode carries the load current does not commutate current and dissipates almost nothing.

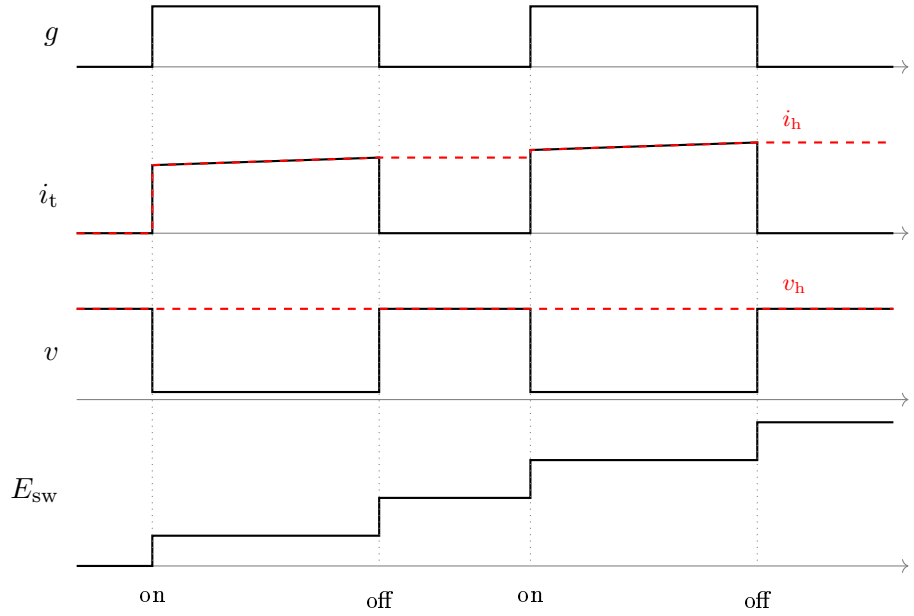


Figure 1: Sampling scheme. The held current i_h freezes at turn-off and is reused at the following turn-on; the held voltage v_h freezes at turn-on and is reused at the following turn-off. Each event adds a step to the cumulative energy E_sw.

4.3 Event equations

The two commutations are written as a single `when`/`elsewhen` chain acting on one event variable E_sw:

$$E_\mathrm{sw} \leftarrow E_\mathrm{sw} + \left(E_\mathrm{on,ref} + E_\mathrm{rr,ref}\right) \frac{|i_\mathrm{h}|}{I_\mathrm{c,ref}} \frac{|v_\mathrm{h}|}{V_\mathrm{dc,ref}} \qquad \text{on } g \uparrow (V_\mathrm{th} + V_\mathrm{hyst}), \tag{7}$$

$$E_\mathrm{sw} \leftarrow E_\mathrm{sw} + E_\mathrm{off,ref} \frac{|i_\mathrm{h}|}{I_\mathrm{c,ref}} \frac{|v_\mathrm{h}|}{V_\mathrm{dc,ref}} \qquad \text{on } g \downarrow (V_\mathrm{th} - V_\mathrm{hyst}). \tag{8}$$

Two separate `when` clauses updating the same event variable are rejected by the compiler, because it cannot prove that the two predicates are mutually exclusive. The `elsewhen` form makes exclusivity structural. The hysteresis band V_hyst around V_th prevents multiple events on a noisy or slow gate edge; the gate waveform must cross both thresholds, otherwise events are missed and the switching loss is under-estimated.

4.4 Attribution of the recovery energy

The reverse recovery energy of a diode is dissipated when the diode is commutated off by the turn-on of the *complementary* switch of the same leg. That instant is not observable from the terminals of a single switch-position model. The model therefore charges E_rr to the turn-on event of its own transistor, together with E_on.

In a two-device leg made of identical instances this preserves the total leg loss exactly and misplaces only its distribution: the pair $E_\mathrm{on} + E_\mathrm{rr}$ appears at the switch that turns on, instead of E_on at that switch and E_rr at the diode of the opposite one. Since the present model already

lumps the IGBT and its FWD onto one thermal port, the additional error concerns only the split between the upper and lower positions of a leg, which is symmetric over a fundamental period. If the diode is modelled as a separate component with its own recovery energy, $E_{\mathrm{rr,ref}}$ must be set to zero here to avoid double counting.

This approximation is acceptable only as long as the two chips share one thermal port. As soon as the junction temperatures are computed separately it becomes a real error, because the recovery energy would be assigned to the diode during the half period in which that diode does not conduct, flattening exactly the peak that the calculation is meant to find. The two-port version therefore detects the recovery event locally; see Section 8.

5 From energy steps to dissipated power

E_{sw} is a staircase. Its derivative is a train of Dirac impulses, which cannot be injected into a thermal network. The model converts the staircase into power with a first-order lag. Let

$$\dot{E}_{\mathrm{lp}} = \frac{E_{\mathrm{sw}} - E_{\mathrm{lp}}}{\tau_{\mathrm{p}}}, \qquad P_{\mathrm{sw}} = \frac{E_{\mathrm{sw}} - E_{\mathrm{lp}}}{\tau_{\mathrm{p}}} = \dot{E}_{\mathrm{lp}}. \tag{9}$$

A step of height ΔE occurring at t_k produces

$$P_{\mathrm{sw}}(t) = \frac{\Delta E}{\tau_{\mathrm{p}}}\, e^{-(t-t_k)/\tau_{\mathrm{p}}}, \qquad \int_{t_k}^{\infty} P_{\mathrm{sw}}\,\mathrm{d}t = \Delta E. \tag{10}$$

The energy is therefore conserved exactly; only its distribution in time is smoothed. Over an interval containing N events,

$$\int_0^T P_{\mathrm{sw}}\,\mathrm{d}t = E_{\mathrm{lp}}(T) - E_{\mathrm{lp}}(0) = \sum_{k=1}^{N} \Delta E_k - \big[E_{\mathrm{sw}}(T) - E_{\mathrm{lp}}(T)\big], \tag{11}$$

and the residual term is bounded by $\tau_{\mathrm{p}} P_{\mathrm{sw}}(T)$, i.e. by the energy of roughly one time constant of switching loss. In steady state the mean of P_{sw} equals $f_{\mathrm{sw}}\big(E_{\mathrm{on}} + E_{\mathrm{off}} + E_{\mathrm{rr}}\big)$, as it should.

The total instantaneous loss delivered to the thermal port is

$$P = \underbrace{v\,i}_{\text{conduction}} + P_{\mathrm{sw}}. \tag{12}$$

5.1 Choice of τ_{p}

τ_{p} must satisfy two conflicting requirements. It should span several switching periods, so that the power seen by the thermal network is a smooth average rather than a sequence of spikes, and it should stay below the smallest thermal time constant that is to be resolved. For the reference module the fastest Foster stage has $\tau_1 = 19.6\,\mu\mathrm{s}$, which is shorter than a switching period at any realistic f_{sw} for a 1200 A device: the junction temperature ripple inside a switching period is not resolvable with averaged losses in any case, and τ_{p} of a few switching periods is the appropriate choice. The default 200 µs corresponds to a few periods at $f_{\mathrm{sw}} = 2\,\mathrm{kHz}$ to 5 kHz. Junction temperature swings at the output fundamental frequency, which are usually the quantity of interest, are unaffected as long as $\tau_{\mathrm{p}} \ll 1/f_{\mathrm{out}}$.

6 Thermal port

The thermal port is declared as a conserving node and its heat flow is bound in the `branches` section:

```
branches
    i : p.i -> n.i;
    Q : H.Q -> *;
end
```

The balancing variable `H.Q` may only be referenced in `branches`; using it directly in `equations` is a compilation error. With the declaration above, Q is positive when heat flows *into* the component, so the dissipation is imposed as

$$Q = -P. \tag{13}$$

The equivalent form `Q : * -> H.Q` together with $Q = +P$ is also valid.

The component contains no thermal capacitance and no thermal resistance: it is a pure heat source. The junction-to-case impedance must be attached externally. For the reference module the datasheet gives a four-stage Foster network in normalised form, which becomes, for the IGBT ($R_{\mathrm{th(j-c)Q}} = 28\,\mathrm{K/kW}$):

Table 3: Foster network derived from the normalised transient thermal impedance data, IGBT chip.

k	r_k (–)	R_k (mK/W)	τ_k (s)	C_k (J/K)
1	0.0124	0.347	0.000 019 6	0.056
2	0.0739	2.069	0.001 40	0.677
3	0.3510	9.828	0.0179	1.821
4	0.5630	15.764	0.0944	5.988

with $R_k = r_k R_{\mathrm{th(j-c)Q}}$ and $C_k = \tau_k / R_k$. The same normalised coefficients apply to the FWD with $R_{\mathrm{th(j-c)D}} = 43\,\mathrm{K/kW}$. Because the single-port model has one thermal port, one network must be chosen for the pair; using the IGBT network is the usual choice when the IGBT carries the larger share of the loss. The two-port version uses both networks, one per port.

7 Parameter extraction from the datasheet

Two points deserve attention when the parameters are taken from a datasheet.

Temperature. The tabulated switching energies of the reference module are specified at $T_{\mathrm{vj}} = 150\,^{\circ}\mathrm{C}$, whereas the conduction voltages are tabulated at 25 °C, 125 °C and 150 °C. A parameter set that is consistent at 125 °C therefore requires reading the energies from the dashed curves rather than from the table. For the reference module the resulting values at 1200 A are approximately 130 mJ, 285 mJ and 205 mJ for E_{on}, E_{off} and E_{rr}, i.e. some 5 % to 10 % below the tabulated 150 °C figures. These are graphical readings and should be treated as such.

Gate resistance. The tabulated energies are measured at $R_{\mathrm{G}} = 0\,\Omega$, which is not an operating condition. The datasheet curves of switching energy versus external gate resistance show a strong dependence: at the maximum recommended turn-on resistance of $6.8\,\Omega$ the turn-on energy is roughly an order of magnitude larger than at $R_{\mathrm{G}} = 0\,\Omega$, while the recovery energy decreases and the turn-off energy roughly doubles at the maximum recommended turn-off resistance of $15\,\Omega$. Entering the tabulated values without correcting for the actual gate resistance leads to a gross under-estimate of the turn-on loss. The energies must be read at the gate resistance of the actual driver.

8 Variant with separate thermal ports

The component `igbt_simple_2th` keeps the equations of Sections 3–5 unchanged and splits the loss into two independent paths, one per chip. It adds two thermal ports, `Ht` for the transistor and `Hd` for the diode, two power outputs `Pt` and `Pd`, and one parameter, $V_{\mathrm{rr,th}}$ (default 50 V), used to detect the recovery event.

8.1 Loss split

Conduction separates naturally, since the two branch currents are already distinct state variables of the model. The switching energies are accumulated in two event variables, E_{t} for the transistor and E_{d} for the diode, each with its own first-order lag:

$$P_{\mathrm{t}} = v\, i_{\mathrm{t}} + \frac{E_{\mathrm{t}} - E_{\mathrm{lp,t}}}{\tau_{\mathrm{p}}}, \qquad E_{\mathrm{t}} \text{ fed by } E_{\mathrm{on}},\ E_{\mathrm{off}}, \tag{14}$$

$$P_{\mathrm{d}} = v\, i_{\mathrm{d}} + \frac{E_{\mathrm{d}} - E_{\mathrm{lp,d}}}{\tau_{\mathrm{p}}}, \qquad E_{\mathrm{d}} \text{ fed by } E_{\mathrm{rr}}. \tag{15}$$

The two heat flows are bound to the two ports as $Q_{\mathrm{t}} = -P_{\mathrm{t}}$ and $Q_{\mathrm{d}} = -P_{\mathrm{d}}$ with the sign convention of Section 6.

8.2 Local detection of the recovery event

The instant at which the diode is commutated off is imposed by the complementary switch of the leg and is not directly observable from the terminals of a single switch position. It is, however, observable indirectly: it is the only instant at which the terminal voltage rises from the diode conduction level to the blocking level while the transistor is not the cause of the transition. The model therefore triggers on

$$v \uparrow V_{\mathrm{rr,th}}, \qquad V_{\mathrm{f}} < V_{\mathrm{rr,th}} \ll V_{\mathrm{dc}}, \tag{16}$$

and scales E_{rr} with a held diode current i_{dh} governed by a three-branch law:

$$\dot{i}_{\mathrm{dh}} = \begin{cases} \dfrac{|i_{\mathrm{d}}| - i_{\mathrm{dh}}}{\tau_{\mathrm{s}}}, & v < -V_{\mathrm{f}} \quad \text{(diode conducting: track)}, \\ -\dfrac{i_{\mathrm{dh}}}{\tau_{\mathrm{s}}}, & g > V_{\mathrm{th}} \quad \text{(transistor gated on: flush)}, \\ 0, & \text{otherwise} \quad \text{(dead time: hold)}. \end{cases} \tag{17}$$

The flush branch is what makes the detection self-selecting. The voltage rise above $V_{\mathrm{rr,th}}$ occurs once per switching period in both half periods of the output fundamental, but only in one of them does it correspond to a recovery, and (17) guarantees that i_{dh} is non-zero only in that one. Table 4 follows the two cases.

Table 4: Behaviour of the recovery detection over the two half periods of the output current, for one switch position.

	Diode half period	Transistor half period
Chip carrying the load current	FWD	IGBT
Branch of (17) in the on-state	track	flush
i_{dh} during the dead time	held	≈ 0
Cause of the voltage rise	complementary turn-on	own turn-off
Energy added at the event	$E_{\mathrm{rr}}(i_{\mathrm{dh}})$	≈ 0

The hold branch covers the dead time: the gate is already low while the diode is still conducting, and the current must not be lost before the commutation actually happens. The order of the branches matters, because during the freewheeling interval the gate of the switch position is typically high while the current flows in the diode; testing the diode conduction first ensures that this interval is tracked and not flushed.

8.3 Event chain

The three events are written as a single `when`/`elsewhen` chain:

```
events
    when edge(g > (Vth + Vhyst))
        E_t = E_t + Eon_ref *(abs(i_h) /Ic_ref)*(abs(v_h)/Vdc_ref);
    elsewhen edge(g < (Vth - Vhyst))
        E_t = E_t + Eoff_ref*(abs(i_h) /Ic_ref)*(abs(v_h)/Vdc_ref);
    elsewhen edge(v > V_rr_th)
        E_d = E_d + Err_ref *(abs(i_dh)/Ic_ref)*(abs(v_h)/Vdc_ref);
    end
end
```

The chain form is required for the same reason as in the single-port version: separate `when` clauses acting on the same event variable are rejected because their predicates cannot be proved mutually exclusive. In the transistor half period the turn-off predicate and the voltage-rise predicate become true almost simultaneously; whether the solver resolves them in one or in two event iterations is immaterial, since the recovery term is zero there.

8.4 Thermal connection

Each port takes its own junction-to-case network, built from the normalised Foster coefficients of Section 6 scaled with $R_{\mathrm{th(j-c)Q}} = 28\,\mathrm{K/kW}$ for `Ht` and $R_{\mathrm{th(j-c)D}} = 43\,\mathrm{K/kW}$ for `Hd`. Both networks are connected to a common case node, followed by $R_{\mathrm{th(c-s)}} = 10\,\mathrm{K/kW}$ per module and by the heatsink model. The internal nodes of a Foster network have no physical meaning, but its terminal behaviour reproduces the datasheet Z_{th}, and connecting both chips to a shared case node is the usual practice.

8.5 Use in a dimensioning calculation

With the two ports available, the peak junction temperature of each chip becomes a direct simulation output. Two points are worth checking explicitly.

First, the peak depends on the output frequency. At low fundamental frequency the thermal ripple over a half period becomes dominant and is not captured by the average loss: the transient at the lowest fundamental frequency of the duty profile, not the mean value, is the sizing case.

Second, the diode is not necessarily the less critical chip. It carries the full load current during a half period, its junction-to-case resistance is significantly higher than that of the transistor, and its recovery loss is concentrated in the same half period in which it conducts. A FWD with a lower average loss than the IGBT can therefore reach a higher peak junction temperature, which is precisely the effect that the single-port version, and the approximate attribution of E_{rr}, would hide.

9 Validity limits

The model does not represent:

- the switching transient, hence no $\mathrm{d}v/\mathrm{d}t$ or $\mathrm{d}i/\mathrm{d}t$ stress, no overshoot due to stray inductance, no tail current and no recovery current on the electrical terminals;

- the temperature dependence of the conduction drops and of the switching energies;
- the dependence of the switching energies on the gate resistance, the stray inductance and the DC-link decoupling;
- the difference between IGBT and FWD junction temperatures, in the single-port version only;
- any short-circuit, desaturation or avalanche behaviour;
- the non-linearity of the output characteristic, unless R_{on} and R_{d} are set to the datasheet slopes.

Within these limits the model reproduces the loss balance of a hard-switched converter with the accuracy of the datasheet data themselves, at a computational cost close to that of an ideal switch. Validation against a vendor loss calculator, at fixed junction temperature and identical gate resistance, is the recommended acceptance test.

A Source listings

A.1 Single thermal port

```
component igbt_simple
% IGBT_SIMPLE Simplified IGBT + antiparallel FWD with thermal port
%
% Conduction : fixed Vce_sat (IGBT) and Vf (diode); Ron/Rd only for
% numerical conditioning, not for accuracy.
% Switching : single-point Eon/Eoff/Err, linearly scaled with the actual
% commutated current and blocking voltage:
% E = E_ref * (Ic/Ic_ref) * (Vdc/Vdc_ref)
% Thermal : total loss (conduction + switching) is injected into port H.
%
% Ic and Vdc are not taken at the switching event itself (the solver value
% there is already the post-transition one) but from two fast sample&hold
% states:
% i_h : IGBT current, tracked while the gate is ON -> value before turn-off
% and, one period later, the current to be commutated at turn-on
% v_h : terminal voltage, tracked while the gate is OFF -> DC-link voltage
%
% Err is charged to the turn-on event of this device (physically it belongs
% to the FWD of the complementary switch). Leg losses are correct, the
% split between IGBT and FWD junction is approximate. Set Err_ref = 0 if the
% diode is modelled separately.
%
% Default parameters: Mitsubishi CM1200DW-34T, 1700 V / 1200 A.

nodes
    p = foundation.electrical.electrical; % C:left
    n = foundation.electrical.electrical; % E:right
    H = foundation.thermal.thermal; % H:left
end

inputs
    g = { 0, 'V' }; % G:left
end

outputs
    P = { 0, 'W' }; % P:right
end

```

```
parameters
    % --- conduction (Tvj = 125 C, terminal values) ---
    Vce_sat = { 2.40, 'V' }; % Collector-emitter on-state voltage
    Vf = { 2.80, 'V' }; % Diode forward voltage

    % --- gate ---
    Vth = { 6, 'V' }; % Gate-emitter threshold voltage
    Vhyst = { 0.85, 'V' }; % Half hysteresis band for the switching events

    % --- switching, single point ---
    Eon_ref = { 138, 'mJ' }; % Turn-on energy at Ic_ref, Vdc_ref
    Eoff_ref = { 309, 'mJ' }; % Turn-off energy at Ic_ref, Vdc_ref
    Err_ref = { 220, 'mJ' }; % Reverse recovery energy at Ic_ref, Vdc_ref
    Ic_ref = { 1200, 'A' }; % Reference current of the energies
    Vdc_ref = { 1000, 'V' }; % Reference voltage of the energies

    % --- numerical ---
    Ron = { 0.05, 'mOhm' }; % IGBT on-state series resistance
    Rd = { 0.05, 'mOhm' }; % Diode series resistance
    Goff = { 1e-6, '1/Ohm'}; % Off-state conductance
    tau_s = { 1, 'us' }; % Sample&hold time constant (<< on-time)
    tau_p = { 200, 'us' }; % Loss averaging time constant (2-5 x Tsw)
end

variables
    i = { 0, 'A' }; % terminal current, C -> E
    v = { 0, 'V' }; % terminal voltage, C - E
    i_t = { 0, 'A' }; % IGBT current
    i_d = { 0, 'A' }; % diode current
    i_h = { 0, 'A' }; % held commutated current
    v_h = { 0, 'V' }; % held blocking voltage
    E_lp = { 0, 'J' }; % low-pass state of the switching energy
    Q = { 0, 'W' }; % heat flow into the component through H
end

variables (Event = true)
    E_sw = { 0, 'J' }; % cumulative switching energy
end

branches
    i : p.i -> n.i;
    Q : H.Q -> *; % positive Q = heat flowing INTO the component
end

equations
    assert(Vce_sat > 0)
    assert(Vf > 0)
    assert(Ic_ref > 0)
    assert(Vdc_ref > 0)
    assert(tau_s > 0)
    assert(tau_p > 0)
    assert(Vhyst > 0)

    v == p.v - n.v;
    i == i_t + i_d;

    % ---------- conduction ----------
    if (g > Vth) && (v > Vce_sat)
        i_t == (v - Vce_sat)/Ron;
    else
        i_t == v*Goff;
    end

```

```
    if v < -Vf
        i_d == (v + Vf)/Rd;
    else
        i_d == v*Goff;
    end

    % ---------- sample & hold of Ic and Vdc ----------
    if g > Vth
        i_h.der == (i_t - i_h)/tau_s; % track while ON
    else
        i_h.der == 0; % hold while OFF
    end

    if (g < Vth) && (v > 0)
        v_h.der == (v - v_h)/tau_s; % track the blocking voltage
    else
        v_h.der == 0;
    end

    % ---------- losses ----------
    % E_sw is a staircase; the first-order lag turns each step into a
    % power pulse whose time integral is exactly the step height.
    E_lp.der == (E_sw - E_lp)/tau_p;

    P == v*i + (E_sw - E_lp)/tau_p;
    Q == -P; % dissipated power leaves through H
end

events
    % A single when/elsewhen chain: the two branches are mutually exclusive by
    % construction, so E_sw is updated by one event equation only.
    % turn-on: Eon of this IGBT + Err of the complementary FWD
    when edge(g > (Vth + Vhyst))
        E_sw = E_sw + (Eon_ref + Err_ref)*(abs(i_h)/Ic_ref)*(abs(v_h)/Vdc_ref);

    % turn-off
    elsewhen edge(g < (Vth - Vhyst))
        E_sw = E_sw + Eoff_ref*(abs(i_h)/Ic_ref)*(abs(v_h)/Vdc_ref);
    end
end

end
```

A.2 Separate thermal ports

```
component igbt_simple_2th
% IGBT_SIMPLE_2TH Simplified IGBT + antiparallel FWD, separate thermal ports
%
% Same modelling assumptions as IGBT_SIMPLE, but the losses of the transistor
% and of the diode are kept separate and delivered to two thermal ports:
% Ht : IGBT chip -> conduction (v*i_t) + Eon + Eoff
% Hd : FWD chip -> conduction (v*i_d) + Err
%
% Err attribution
% ---------------
% The recovery energy of THIS diode is dissipated when the complementary
% switch of the leg turns on. That instant is detected locally as the moment
% the terminal voltage rises above V_rr_th while the diode was conducting.
% The diode current before the commutation is kept in the state i_dh, which
% tracks the diode current while it conducts, is flushed to zero while the
% transistor is gated on, and holds otherwise (dead time). As a result:
% - in the half period in which the diode conducts, one Err event per
```

```
% switching period is generated with the correct current;
% - in the half period in which the transistor conducts, the same voltage
% rise occurs at turn-off but i_dh has been flushed, so Err = 0.
%
% Default parameters: Mitsubishi CM1200DW-34T, 1700 V / 1200 A.
% Suggested thermal networks (datasheet, normalised Foster coefficients):
% Ht : Rth(j-c)Q = 28 K/kW Hd : Rth(j-c)D = 43 K/kW
% both connected to a common case node, then Rth(c-s) = 10 K/kW per module.

nodes
    p = foundation.electrical.electrical; % C:left
    n = foundation.electrical.electrical; % E:right
    Ht = foundation.thermal.thermal; % Ht:left
    Hd = foundation.thermal.thermal; % Hd:right
end

inputs
    g = { 0, 'V' }; % G:left
end

outputs
    Pt = { 0, 'W' }; % Pt:right
    Pd = { 0, 'W' }; % Pd:right
end

parameters
    % --- conduction (Tvj = 125 C, terminal values) ---
    Vce_sat = { 2.40, 'V' }; % Collector-emitter on-state voltage
    Vf = { 2.80, 'V' }; % Diode forward voltage

    % --- gate ---
    Vth = { 6, 'V' }; % Gate-emitter threshold voltage
    Vhyst = { 0.85, 'V' }; % Half hysteresis band for the switching events

    % --- switching, single point ---
    Eon_ref = { 138, 'mJ' }; % Turn-on energy at Ic_ref, Vdc_ref
    Eoff_ref = { 309, 'mJ' }; % Turn-off energy at Ic_ref, Vdc_ref
    Err_ref = { 220, 'mJ' }; % Reverse recovery energy at Ic_ref, Vdc_ref
    Ic_ref = { 1200, 'A' }; % Reference current of the energies
    Vdc_ref = { 1000, 'V' }; % Reference voltage of the energies

    % --- numerical ---
    V_rr_th = { 50, 'V' }; % Blocking detection level for the Err event
    Ron = { 0.05, 'mOhm' }; % IGBT on-state series resistance
    Rd = { 0.05, 'mOhm' }; % Diode series resistance
    Goff = { 1e-6, '1/Ohm'}; % Off-state conductance
    tau_s = { 1, 'us' }; % Sample&hold time constant (<< on-time)
    tau_p = { 200, 'us' }; % Loss averaging time constant (2-5 x Tsw)
end

variables
    i = { 0, 'A' }; % terminal current, C -> E
    v = { 0, 'V' }; % terminal voltage, C - E
    i_t = { 0, 'A' }; % IGBT current
    i_d = { 0, 'A' }; % diode current
    i_h = { 0, 'A' }; % held IGBT commutated current
    i_dh = { 0, 'A' }; % held diode current before recovery
    v_h = { 0, 'V' }; % held blocking voltage
    E_lpt = { 0, 'J' }; % low-pass state, IGBT switching energy
    E_lpd = { 0, 'J' }; % low-pass state, diode switching energy
    Qt = { 0, 'W' }; % heat flow into the component through Ht
    Qd = { 0, 'W' }; % heat flow into the component through Hd
end
```

```
81
82  variables (Event = true)
83      E_t = { 0, 'J' }; % cumulative IGBT switching energy
84      E_d = { 0, 'J' }; % cumulative diode recovery energy
85  end
86
87  branches
88      i : p.i -> n.i;
89      Qt : Ht.Q -> *; % positive Q = heat flowing INTO the component
90      Qd : Hd.Q -> *;
91  end
92
93  equations
94      assert(Vce_sat > 0)
95      assert(Vf > 0)
96      assert(Ic_ref > 0)
97      assert(Vdc_ref > 0)
98      assert(tau_s > 0)
99      assert(tau_p > 0)
100     assert(Vhyst > 0)
101     assert(V_rr_th > Vf)
102
103     v == p.v - n.v;
104     i == i_t + i_d;
105
106     % ---------- conduction ----------
107     if (g > Vth) && (v > Vce_sat)
108         i_t == (v - Vce_sat)/Ron;
109     else
110         i_t == v*Goff;
111     end
112
113     if v < -Vf
114         i_d == (v + Vf)/Rd;
115     else
116         i_d == v*Goff;
117     end
118
119     % ---------- sample & hold ----------
120     % IGBT current: tracked while ON, held while OFF.
121     if g > Vth
122         i_h.der == (i_t - i_h)/tau_s;
123     else
124         i_h.der == 0;
125     end
126
127     % Blocking voltage: tracked while OFF and forward biased.
128     if (g < Vth) && (v > 0)
129         v_h.der == (v - v_h)/tau_s;
130     else
131         v_h.der == 0;
132     end
133
134     % Diode current magnitude: tracked while the diode conducts, flushed while
135     % the transistor is gated on, held during the dead time.
136     if v < -Vf
137         i_dh.der == (-i_d - i_dh)/tau_s;
138     elseif g > Vth
139         i_dh.der == -i_dh/tau_s;
140     else
141         i_dh.der == 0;
142     end
143
```

```
144     % ---------- losses ----------
145     E_lpt.der == (E_t - E_lpt)/tau_p;
146     E_lpd.der == (E_d - E_lpd)/tau_p;
147 
148     Pt == v*i_t + (E_t - E_lpt)/tau_p;
149     Pd == v*i_d + (E_d - E_lpd)/tau_p;
150 
151     Qt == -Pt;
152     Qd == -Pd;
153 end
154 
155 events
156     % One when/elsewhen chain: the branches are mutually exclusive by
157     % construction, so no event variable is updated by two event equations.
158 
159     % turn-on of the IGBT
160     when edge(g > (Vth + Vhyst))
161         E_t = E_t + Eon_ref*(abs(i_h)/Ic_ref)*(abs(v_h)/Vdc_ref);
162 
163     % turn-off of the IGBT
164     elsewhen edge(g < (Vth - Vhyst))
165         E_t = E_t + Eoff_ref*(abs(i_h)/Ic_ref)*(abs(v_h)/Vdc_ref);
166 
167     % reverse recovery of the FWD, forced by the complementary switch
168     elsewhen edge(v > V_rr_th)
169         E_d = E_d + Err_ref*(abs(i_dh)/Ic_ref)*(abs(v_h)/Vdc_ref);
170     end
171 end
172 
173 end
```